Assigment No. 4

Time complexity & space complexity.

1)

```
1
1 2
1 2 3
1 2 3 4
1 2 3
1 2
1
```

1st half,

```
for(int i=1 ;i<=n ;i++){
    for(int j=1 ;j<=n-i ;j++){
        Sop(" ");
    }
    for(int j=1 ;j<=i ;j++){
        Sop(i+" ");
    }
}
```

2nd half.

```
for(int i=n-1 ;i>=1 ;i--){
    for(int j=1 ;j<=n-i ;j++){
        Sop(" ");
    }
    for(int j=1 ;j<=i ;j++){
        Sop(i+" ");
    }
}
```

outer loop

i) 1 to n → n times

inner loop

1st j loop = 1 to n-i

≈ n times

2nd j loop = 1 to i

≈ n

$n \times n = 2n^2$ ~~Avoid~~ Avoid constant

second outer loop

n-1 to 1 ≈ n

and triangle or space

≈ n

$n \times n = n^2$

$n^2 + n^2 = 2n^2$ Avoid constan

Time C ⇒ n^2

use constan space

i, j, n

∴ O(1)

Time complexity = $O(n^2)$

Space complexity = $O(1)$

2)

```
* * * *
*   *
*   *
* * * *
```

```
for (int i=1 ; i<=n; i++){
    for(int j=1 ; j<=n ; j++){
        if( i==1 || i==n || j==1 || j=
            Sop("*");
        }else{
            Sop(" ");
        }
    }
```

Outer loop (i)
1 to n = n times
innerloop (j)
1 to n = n times.

$\therefore n \times n = n^2$

Time Complexity = $O(n^2)$
Space Complexity = $O(1)$

3)

```
*
* *
* * *
* * * *
* * * * *
```

```
for(int i=1 ; i<=n ; i++){
    for(int j=1 ; j<=i ; j++){
        Sop("*");
    }
    Sopln();
```

$i=1 \quad j=1$
$i=2 \quad j=2$
$i=3 \quad j=3$
$i=n \quad j=n$

} times loop फिरेगा.

$\therefore \frac{n(n+1)}{2} = \frac{n^2+n}{2} = \frac{n^2}{2}$ worst case

$= n^2$

TC = $O(n^2)$
SC = $O(1)$

4)

```
1        1
  2    2
    3
  4    4
5        5
```

```
for(int i=1; i<=n; i++){
    for(int j=1; j<=n; j++){
        if(i==j || i+j==n+1){
            cout("*");
        }else{
            cout(" ");
        }
    }
}
```

i loop = 1 to n
j loop = 1 to n } $n \times n = n^2$

Time complexity = $O(n^2)$
Space complexity = $O(1)$

5)

```
1           1
12         21
123       321
1234 4321
12344321
123       321
12         21
1           1
```

```
for(int i=1; i<=n; i++){
    for(int j=1; j<=i; j++){
        cout("*" j);
    }
}
for(int j=1; j<=n(n-1); j++){
    cout(" ");
}
for(int j=i; j>=1; j--){
    cout(j);
}
```

1st half 1 3
i loop → 1 to n = n 1 2nd half
2 triangle → 1 to i = n 1
1 Space → 1 to 2(n-1) = n 1

$\therefore n \times (n+n) = 2n^2$
$\therefore n^2$

```
for(int i=n; i>=1; i--){
    for(int j=1; j<=i; j++){
        cout(j);
    }
}
```

2nd half

i = loop → n-1 to 1 = n

2 triangloop = 1 to i = n

1 space loop = 1 to n = n

∴ n(n+n) = n(2n)

= $2n^2$ Avoid constant

= n^2

```
for(int j=1; j<=(2*(n-i)); j
        Sop(" ");
}
for(int j=i; j>=1; j
        Sop(j);
}
```

∴ 1st + 2nd half

$n^2 + n^2$

$= 2n^2$

$= n^2$

Time complexity = $O(n^2)$

Space complexity = $O(1)$